

UTSAV CHAUDHARY

Texas | utsavmaan28@gmail.com | (607) 296-9583 | thatinsaneguy.com | in/utsavmaan28 | github/UttU28

Senior **Data GenAI Engineer** with **5+ years** building **Python** data pipelines, **ML** models, and **GenAI** systems for analytics, internal tools, and tokenization workflows. Proficient in **multi-agent** orchestration with **GPT-4o** and **GPT-4o-mini**, **RAG** (chunking, **embeddings**, vector search, top-k retrieval), **Hugging Face** fine-tunes, and **FastAPI** / **Node.js** service layers. Experienced with **ETL** and time-series **ML** (**SARIMA**, **Pandas**), plus **Ethereum** token and wallet work (**ERC-20**, **ERC-1404**, **ERC-3643**, **Fireblocks**, **MetaMask**).

Technical Skills

Programming & Frameworks: Python, JavaScript, React.js, Node.js, FastAPI, Flask, Pydantic, REST APIs, WebSockets, HTML, CSS, C#

AI, ML & GenAI: GPT-4o, GPT-4o-mini, multi-agent systems, Hugging Face, Transformers, RAG, embeddings, vector databases, cosine similarity, ANN, top-k retrieval, SARIMA, Pandas, NumPy, TensorFlow Lite

Data & Analytics: ETL, data ingestion, chunking, metadata tagging, D3.js, Chart.js, SQL, MAPE, RMSE

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis, Firebase, Firestore, Amazon RDS, Amazon Aurora, Amazon DynamoDB, Amazon DocumentDB, Amazon ElastiCache, Azure SQL, vector stores

Blockchain & Web3: Solidity, Ethereum, ERC-20, ERC-1404, ERC-3643, Fireblocks, MetaMask

Tools & APIs: Selenium, BeautifulSoup, Google Search API, Gmail API, Unity3D, Git, GitHub, Bitbucket, Jira, Linux, Windows, macOS

Experience

Sr. Python Blockchain Developer, Hybrid, **LABS196 Innovations**, TX

Feb 2025 - Present

- Partnered with product and ops to design a **multi-agent** Python workflow. I owned the **Manager agent (GPT-4o)** that plans tasks, picks tools, and routes work to specialist agents.
- Worked with the team on an **Event Finder agent (GPT-4o-mini)**. I wired **Google Search API** and **Selenium** tools so it could scrape pages and hand structured event lists back to the manager.
- Built the **Data Refinement agent** (DB and **data cleaner** tools) while coordinating with backend on **MongoDB/SQL** schemas so cleaned records were usable by later agents.
- Collaborated with outreach on an **Email Formation agent (GPT-4o)** and **Gmail API** drafts. I added manager approval gates so nothing sent without a review step.
- Owned the shared tool layer (search, scrape, DB, cleaner, Gmail) as callable functions with schemas, retries, and logs so the rest of the team could add tools without rewriting agent prompts.
- Fine-tuned a **Hugging Face** transformer for field extraction (names, dates, venues, status) from messy scrape and CRM text, then plugged it in as a refinement-agent tool before data hit MongoDB.
- Wrote **Solidity** token POCs on **Ethereum** for **ERC-20**, **ERC-1404**, and **ERC-3643** with engineering and compliance review on mint/burn, transfer rules, and event schemas.
- Built the **Web3** wallet and transaction path with **Fireblocks** and **MetaMask** (create, sign, transfer) and used **Python** to parse on-chain activity into datasets the team used for monitoring.

Data Science Specialist, On-Site, **Finix196 Capital**, TX

Jan 2024 - Feb 2025

- Worked with internal support and document owners on a **RAG** QnA tool for finance and tokenization docs. I owned the retrieval pipeline so the team could ask questions instead of digging through folders.
- Wrote **Python** ingestion jobs (chunking, metadata, **embeddings**, **vector DB** upserts) and aligned doc types and tags with the people who owned the source PDFs.
- Implemented query flow (embed question, **cosine similarity** / ANN, **top-k**, **LLM** answer with citations) as the core DS piece, then reviewed sample answers with support before we widened access.
- Automated document refresh in **Python** (new files, re-chunk, re-embed, upsert) so ops did not have to rebuild the index by hand when policies changed.
- Built the chat **UI** in **Node.js** with a **Python/FastAPI** backend, pairing with another engineer on the frontend while I owned retrieval, ranking, and answer generation.
- Tuned chunk size, overlap, metadata filters, and top-k with support feedback so answers stayed on-policy and fewer tickets came back as "wrong doc."

Data Engineer CO-OP, On-Site, **Midh Technologies Inc**, TX

June 2023 - Dec 2023

- Worked on a team **POC** as a co-op to forecast **fuel distribution** by location using **SARIMA**, so planners could match supply to seasonal demand instead of a single average for every site.
- Cleaned and joined location time series in **Python (Pandas)** with seniors: busy vs remote sites, weekend vs weekday load, and seasonal peaks, then split train/test for model checks.
- Helped fit and compare **SARIMA** models per location (order, seasonal period, stationarity) and built a simple predictor that output expected fuel volume for upcoming weeks.
- Pair-programmed with the data science team on error review (MAPE/RMSE by site type) and documented which locations needed their own seasonal pattern vs a shared regional model.
- Presented POC results to ops and mentors, and wrapped the predictor in a small **Python** script/notebook so the team could rerun forecasts when new weekly data landed.

Automation Developer, Hybrid, **Compendious Medialabs Pvt. Ltd.**, IN

June 2021 - July 2022

- Worked with analysts on a **React** and **FastAPI** metrics platform. I owned the **Python** APIs (**Pydantic**, **REST**) that kept analytics calls around **250ms**.
- Built **Python** scrape and **ETL** jobs with **Selenium**, **BeautifulSoup**, and **Pandas**, cutting manual collection time by about **60%**.
- Added **WebSocket** streaming for high-frequency metrics so ops dashboards stayed current without full page reloads.
- Partnered on **D3.js** / **Chart.js** dashboard views and tightened **SQL** curation so reporting was faster to load (about **45%** quicker viz for the same workloads).

Software Engineer, Remote, **VROKCs**, KS

March 2020 - Sept 2020

- Partnered with ML engineers on a **sign language** translation pipeline. I handled **Python** preprocessing for data collection, annotation, and dataset prep.
- Connected that pipeline to **Unity3D** for live ASL in AR, and helped run stakeholder demos with the ML group.
- Built a **Flask** app (**Python**, **HTML**, **CSS**, **JavaScript**) with **TensorFlow Lite** so the team could test the model in the browser via image upload and webcam.
- Tuned inference latency and basic data-quality checks so in-browser predictions stayed usable during those demos.

Game Developer / Web Backend, On-Site, **underDOGS Studio**, IN

March 2019 - July 2019

- Worked with game and backend folks to wire the **Quarters API** across titles (wallets, points-to-Quarters, in-game currency). I owned the conversion and transaction API path.
- Built **POC** modules in **Unity3D** that linked **payments**, **inventory**, and **shipping** into one transaction layer for the team.
- Added unit and integration tests around those APIs with the team before production releases.
- Tightened real-time client/server flows in **Python** and **C#** so payments still behaved when a request failed or the network dropped.

Education

State University of New York at Binghamton

Master of Science in Computer Science

Aug 2022 - Dec 2023

3.44/4.00 GPA

Certifications



Microsoft Certified: DevOps Engineer Expert

AZ-400: Designing and Implementing Microsoft DevOps Solutions

AZ-900: Microsoft Certified: Azure Fundamentals

Google IT Automation with Python (Specialization)

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